6

Introduction to Hypothesis Testing: Inferences About a Single Mean

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Key Concepts

1. Null or statistical hypothesis versus scientific or research hypothesis
2. Type I error (α), type II error (β), and power ($1 - \beta$)
3. Two-tailed versus one-tailed alternative hypotheses
4. Critical regions and critical values
5. *z* test statistic
6. Confidence interval around the mean
7. *t* test statistic
8. *t* distribution, degrees of freedom, and table of *t* distributions

In Chapter 5 we began to move into the realm of inferential statistics. There we considered the following general topics: probability, sampling, and estimation. In this chapter we move totally into the domain of inferential statistics, where the concepts involved in probability, sampling, and estimation can be implemented. The overarching theme of the chapter is the use of a statistical test to make inferences about a single mean. In order to properly cover this inferential test, a number of basic foundational concepts are described in this chapter. Many of these concepts are utilized throughout the remainder of this text. Thus, even though there are likely lots of new concepts introduced in this chapter, a large portion of them will resurface in the remaining chapters (i.e., you'll be able to continue to apply what you learn in this chapter).

The topics described in the chapter include the following: types of hypotheses; types of decision errors; level of significance (α); overview of steps in the decision-making process; inferences about μ when σ is known; Type II error (β) and power ($1 - \beta$); statistical versus practical significance; and inferences about μ when σ is unknown. Concepts to be discussed include the following: the null or statistical hypothesis versus the scientific or research hypothesis; Type I error (α), Type II error (β), and power ($1 - \beta$); two-tailed versus one-tailed alternative hypotheses; critical regions and critical values; the z test statistic; the confidence interval around the mean; the t test statistic; and the t distribution, degrees of freedom, and table of t distributions. Our objectives are that by the end of this chapter, you will be able to (a) understand the basic concepts of hypothesis testing; (b) utilize the normal and t tables; and (c) understand, determine, and interpret the results from the z test, t test, and confidence interval procedures.

6.1 Inferences About a Single Mean and How They Work

6.1.1 Characteristics

You may remember Ott Lier and his graduate student colleagues from previous chapters as they have assisted in solving various statistical dilemmas. We see Ott has now been tasked with quite an interesting project.

Ott Lier has greatly enjoyed working with his colleagues on various statistical projects in which they have been involved through the stats lab. Ott and his group completed their first tasks as research assistants—determining a number of descriptive statistics on data. The faculty advisor for the statistical lab has been contacted by a community partner, Coach Wesley, the local hockey coach, who is interested in examining team skating performance. Ott has been assigned to the project. After consulting with Coach Wesley, Ott determines the most appropriate research question to be the following: *Is the mean skating speed of the hockey team different from the league mean speed of 12 seconds?* Ott suggests a one-sample test of means as the test of inference. His task is to assist Coach Wesley in generating the test of inference to answer his research question.

6.1.1.1 Types of Hypotheses

Hypothesis testing is a decision-making process where two possible decisions are weighed in a statistical fashion. In a way, this is much like any other decision involving two possibilities, such as whether to carry an umbrella with you today or not. In statistical

BOX 6.1 Evidence for Testing the Assumption of Normality

Evidence	Interpretation for Providing Evidence of Normality
Boxplot	Normality suggested when the quartiles are relatively evenly distributed with no outliers.
Histogram	Normality suggested with a relatively bell-shaped curve.
Skewness	Values within an absolute value of 2.0 suggest evidence of normality.
Kurtosis	Values within an absolute value of 2.0 suggest evidence of normality.
Standardized skew and standardized kurtosis	Divide the skew and kurtosis values by their standard errors to get standardized skew and kurtosis values. Compare those values to a critical value (e.g., ±1.65 if $\alpha = .10$; ±1.96 if $\alpha = .05$; ±2.06 if $\alpha = .01$). Standardized skew and kurtosis that are less than the critical value suggest evidence of normality.
D'Agostino's test	Tests the null hypothesis that skewness equals zero, with a statistically significant D'Agostino's test indicating that there is statistically significant skewness.
Bonett-Seier test for Geary's kurtosis	Tests the null hypothesis that data should have a Geary's kurtosis value equal to $\sqrt{2/\pi} = .7979$. A statistically significant test indicates that there is statistically significant kurtosis.
Quantile-quantile (Q-Q) plots	Plots that depict quantiles of the sample distribution to quantiles of the theoretical normal distribution. Points that fall on or closely to the diagonal line of the Q-Q plot suggest evidence of normality.
Detrended quantile-quantile plot	Evidence of normality is provided when the points exhibit little or no pattern around zero (the horizontal line).
Kolmogorov-Smirnov (K-S) and Shapiro-Wilk (S-W) tests	K-S and S-W are formal tests of normality. K-S is conservative; S-W test is usually considered more powerful and is recommended for use with small sample sizes ($n < 50$). Non-statistically significant K-S and S-W results are interpreted to say that our distribution is *not* statistically significantly different than a normal distribution.

6.2 Computing Inferences About a Single Mean Using SPSS

Here we consider what SPSS has to offer in the way of testing hypotheses about a single mean. As with most statistical software, the *t* test is included as an option in SPSS, but the *z* test is not. Thus, instructions for determining the one-sample *t* test using SPSS are presented first.

Step 1. To conduct the one-sample *t* test, go to "Analyze" in the top pulldown menu, then select "Compare Means," and then select "One-Sample T Test." Following the steps in the screenshot shown in Figure 6.8 produces the "One-Sample T Test" dialog box.

6.3 Computing Inferences About a Single Mean Using R

Next we consider **R** for the one-sample *t* test. The scripts are provided within the blocks with additional annotation to assist in understanding how the commands work. Should you want to write reminder notes and annotation to yourself as you write the commands in **R** (and we highly encourage doing so), remember that any text that follows a hashtag (i.e., #) is annotation only and not part of the **R** script. Thus, you can write annotations directly into **R** with hashtags. We encourage this practice so that when you call up the commands in the future, you'll understand what the various lines of code are doing. You may think you'll remember what you did. However, trust us. There is a good chance that you won't. Thus, consider it best practice when using **R** to annotate heavily!

6.3.1 Reading Data into R

```
getwd()
```

R is always pointed to a directory on your computer. To find out which directory it is pointed to, run this "get working directory" command. We will assume that we need to change the working directory, and will use the next line of code to set the working directory to the desired path.

```
setwd("E:/Folder")
```

To set the working directory, use the *setwd* function and change what is in quotation marks here to your file location. Also, if you are copying the directory name, it will copy in slashes. You will need to change the slash (i.e., \) to forward slash (i.e., /). Note that you need your destination name within quotation marks in the parentheses.

```
Ch6_skate <- read.csv("Ch6_skate.csv")
```

The *read.csv* function reads your data into **R**. What's to the left of the <- will be what the data will be called in **R**. In this example, we're calling the **R** dataframe "Ch6_skate." What's to the right of the <- tells **R** to find this particular .csv file. In this example, our file is called "Ch6_skate.csv." Make sure the extension (i.e., .csv) is included in your script. Also note that the name of your file should be in quotation marks within the parentheses.

```
names(Ch6_skate)
```

The *names* function will produce a list of variable names for each dataframe as follows. This is a good check to make sure your data have been read in correctly.

FIGURE 6.11
Reading data into **R**.

```
Ch6_onet
```

This script will output the results from our one sample *t* test into the RStudio console. We see our test statistic value, $t = -4.4721$, with 15 degrees of freedom, and a *p* value of < .001 (specifically $p = .0004475$). The 95% confidence interval of the mean is 9.05 to 10.95. The mean of our variable is 10 and is provided in the 'sample estimates' output.

```
       One Sample t-test

data: Ch6_skate$time

t = -4.4721, df = 15, p-value = 0.0004475

alternative hypothesis: true mean is not equal to 12

95 percent confidence interval:
  9.046787 10.953213

sample estimates:
mean of x
       10
```

FIGURE 6.12 (continued)
Generating the One-Sample *t* Test

6.4 Data Screening

Recall that the one-sample *t* test rests on two assumptions: independence of observations and normality. In terms of data screening to examine the extent to which assumptions were met, we will focus on normality, as independence is a matter of sampling method.

6.4.1 Generating Normality Evidence

As alluded to earlier in the chapter, understanding the distributional shape of your variable, specifically the extent to which normality is a reasonable assumption, is important. In earlier chapters, we saw how we could use the Explore tool in SPSS to generate a number of useful descriptive statistics. In conducting our one-sample *t* test, we can again use Explore to examine the extent to which the assumption of normality is met for our sample distribution. As the general steps for accessing Explore from the top toolbar in SPSS have been presented in previous chapters (e.g., Chapter 4), they will not be reiterated here. Thus, we will begin from the main dialog box. We first move the variable of interest to the "Dependent List" box in the main Explore dialog box. Next, click "Plots" in the upper-right corner. Place a checkmark in the boxes for "Normality plots with tests" and also for "Histogram" (see the screenshot in Figure 6.13a).

6.5 Power Using G*Power

In our discussion of power presented earlier in this chapter, we indicated that the sample size to achieve a desired level of power can be determined *a priori* (before the study is conducted) as well as post hoc (after the study is conducted) using statistical software or power tables. One freeware program for calculating power is G*Power (www.psycho.uni-duesseldorf.de/abteilungen/aap/gpower3/) which can be used to compute both *a priori* sample size and post hoc power analyses (among other things). Using the results of the one-sample *t* test just conducted, let us utilize G*Power to first determine the required sample size given various estimated parameters and then compute the post hoc power of our test.

6.5.1 *A Priori* Power

Step 1. As shown in the screenshot for Step 1 in Figure 6.18, several decisions need to be made from the initial G*Power screen. First, the correct test family needs to be selected. In our case, we conducted a one-sample *t* test; therefore, the default selection of "t tests" is the correct test family. Next, we need to select the appropriate statistical test. We use the arrow to toggle to "Means: Difference from constant (one sample case)."

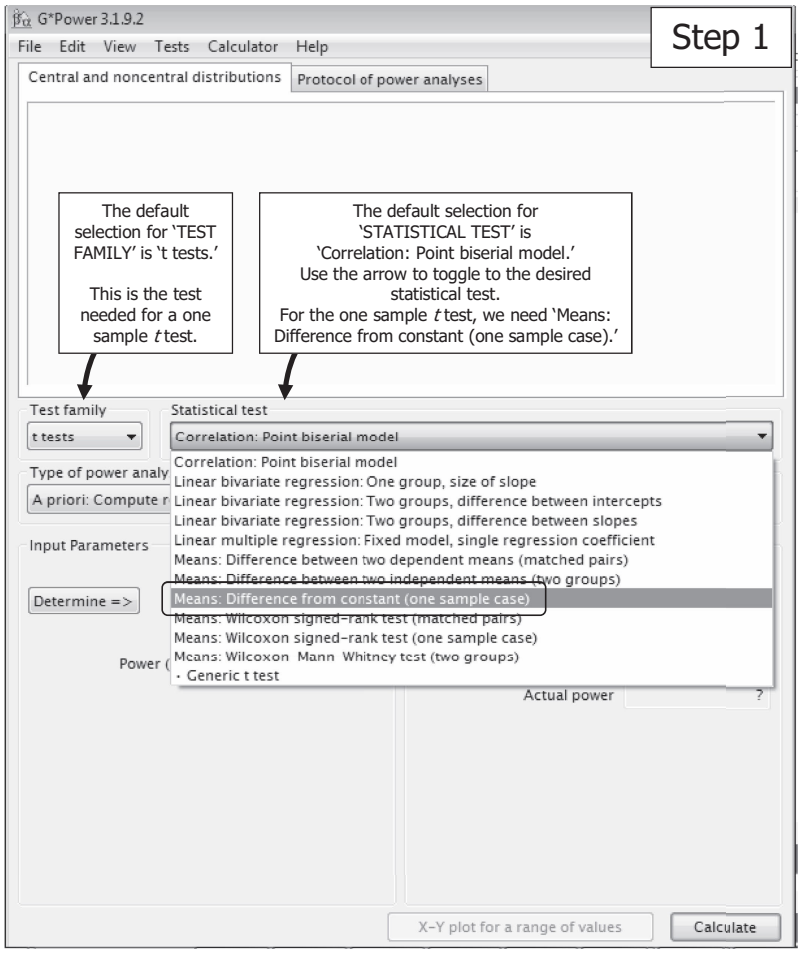


FIGURE 6.18
A priori power: Step 1.

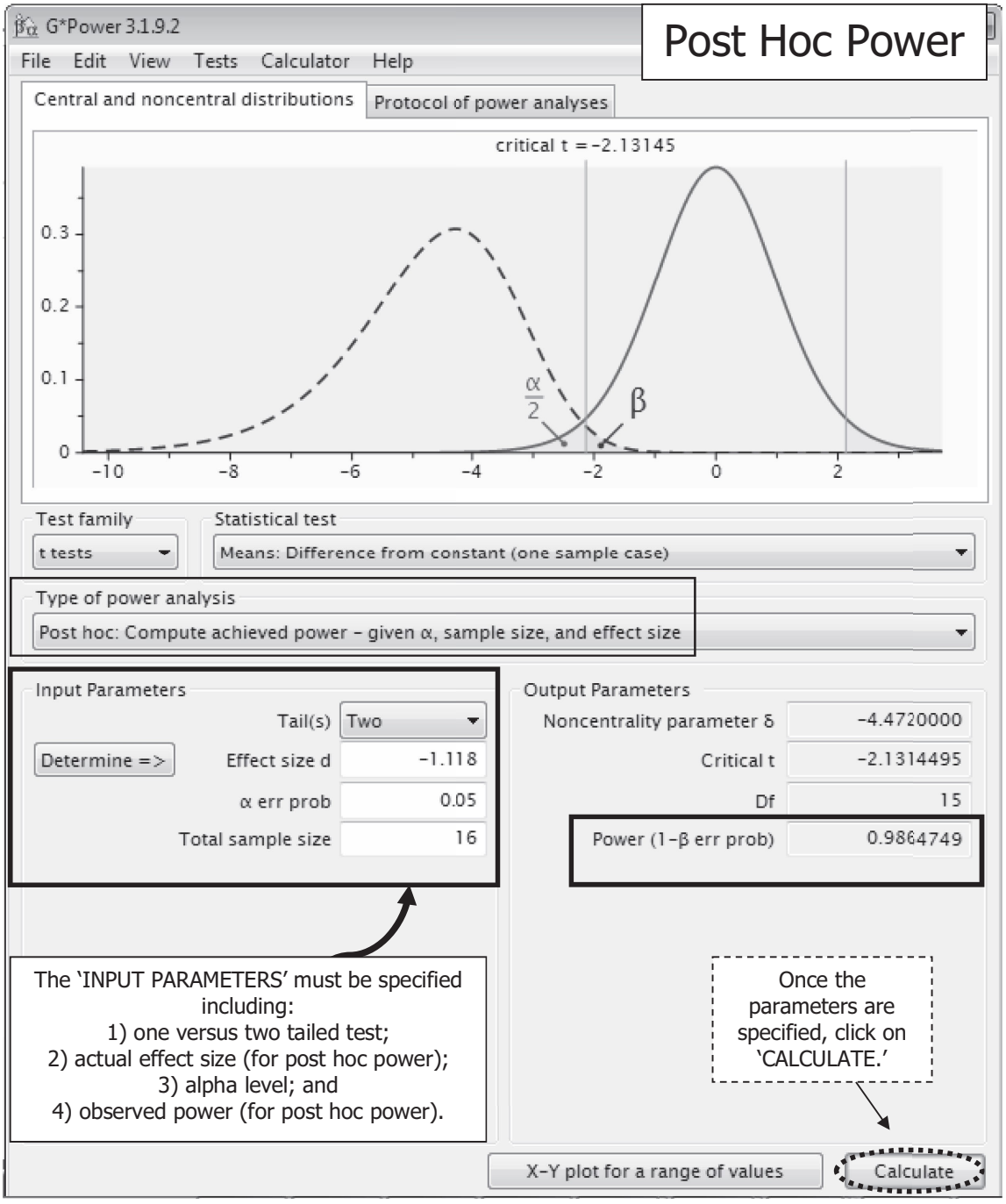


FIGURE 6.23
Post hoc power.

6.6 Research Question Template and Example Write-Up

Let us revisit our graduate research assistant, Ott Lier, who was working with Coach Wesley, a local hockey coach, to assist in analyzing his team's data. As a reminder, Ott's task was to assist Coach Wesley in generating the test of inference to answer the following research question: *Is the mean skating speed of our hockey team different from the league mean*

Remember that for the one-sample mean test, d indicates how many standard deviations the sample mean is from the hypothesized mean. Thus with an effect size of −1.118, there are nearly one and one-quarter standard deviation units between our sample mean and the hypothesized mean. The negative sign simply indicates that our sample mean was the smaller mean (as it is the first value in the numerator of the formula). In this particular example, the negative effect is desired as it suggests the team's average skating time is quicker than the league mean. Using Uanhoro's online calculator (Uanhoro, 2017), we find the confidence interval for the effect size of (−1.7371,−0.4765).

Here is an example APA-style paragraph of results for the skating data (remember that this will be prefaced by the paragraph reporting the extent to which the assumptions of the test were met).

A one sample t test was conducted at an alpha level of .05 to answer the research question: *Is the mean skating speed of a hockey team different from the league mean speed of 12 seconds?* The null hypothesis stated that the team mean speed would not differ from the league mean speed of 12. The alternative hypothesis stated that the team average speed would differ from the league mean. Based on a random sample of 16 skaters, there was a mean time of 10 seconds and a standard deviation of 1.7889 seconds. When compared against the hypothesized mean of 12 seconds, the one-sample t test was shown to be statistically significant ($t = -4.472$, $df = 15$, $p < .001$). Therefore, the null hypothesis that the team average time would be 12 seconds was rejected. This provides evidence to suggest that the sample mean skating time for this particular team was statistically different from the hypothesized mean skating time of the league. Additionally, the effect size d was −1.118 (CI −1.7371,−0.4765), generally interpreted as a large effect (Cohen, 1988), and indicating that there is more than a one standard deviation difference between the team and league mean skating times, with the team speed quicker than the league speed. The post hoc power of the test, given the sample size, two-tailed test, alpha level, and observed effect size, was .986.

6.7 Additional Resources

A number of resources are available for learning more about statistics and how to interpret statistics. In addition to those already cited, Huck (2000) is an excellent general resource to assist in learning more about statistics and how to interpret statistics

Problems

Conceptual Problems

1. In hypothesis testing, the probability of failing to reject H_0 when H_0 is false is denoted by which of the following?

a. α

b. $1 - \alpha$